4 In a binary star system, two stars, each of equal mass 3.5×10^{30} kg, rotate about their common centre of mass O which is equidistant from the centres of the stars, as shown in Fig. 4.1. The separation between the centres of the two stars is 2.0×10^{11} m.

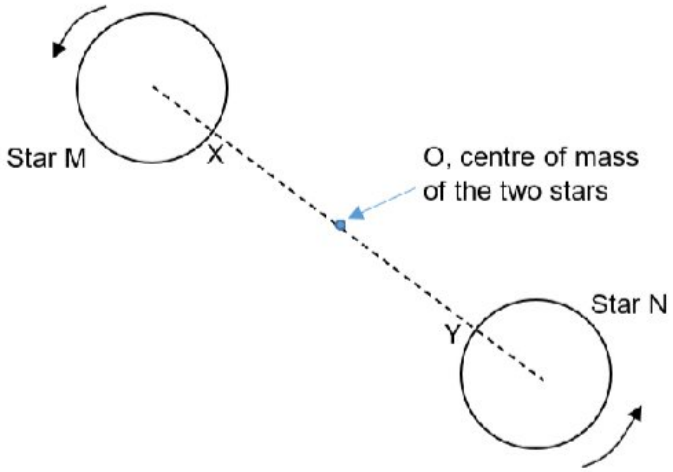


Fig. 4.1

(a) Define gravitational potential at a point.

.....

.....

..... [2]

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(b) Calculate the gravitational potential at O, the centre of mass of the binary star system.

gravitational potential = ... J kg^{-1} [2]

(c) An asteroid passes through point O, at a speed v.

Determine the minimum speed v the asteroid must have if it is to escape from the gravitational pull of the binary star system.

minimum speed v = ... m s^{-1} [2]

(d) On Fig. 4.2, sketch a graph showing the variation with distance of gravitational potential along the line XY between the two stars. Numerical values are not required. [2]

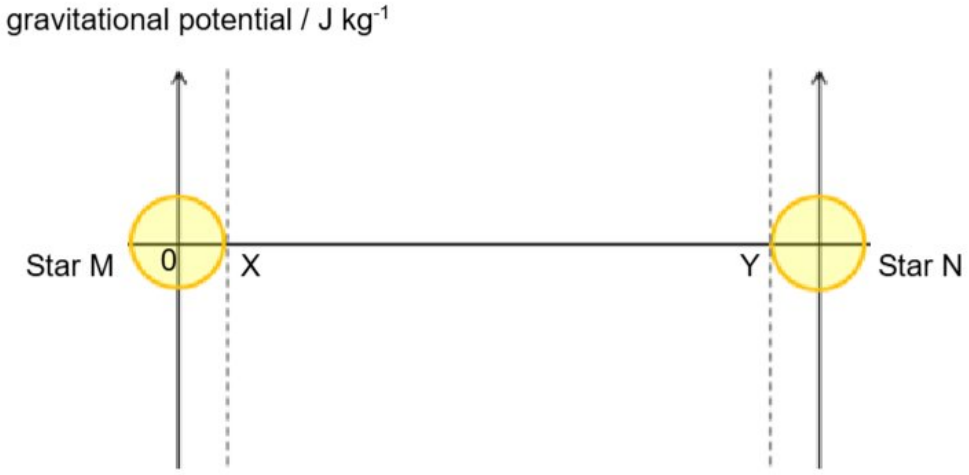


Fig. 4.2

(e) Calculate the time taken for the stars to make one complete revolution about their common centre of mass.

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time = ... s [2]

[Total :10]